Social Deduction Mysteries

The Game Premise

A group of covert operatives are conducting secret investigations. However, some of them are traitors, who will sabotage the investigation to protect their allies. If enough investigations are sabotaged, the public loses faith in the agency and the traitors win. If enough investigations are successful, the traitors lose.

Rules

- 5-9 players
    - For 5-7 player games, there will be one traitor
    - For 8-9 player games, there will be two traitors
    - Roles are randomly and secretly assigned at the beginning

- Every player must take part in at least two investigations (otherwise the traitors automatically win, claiming the agency unfairly distributes work)

- No investigation lineup can be repeated

- Traitors can either sabotage or play along with an investigation. Regular agents cannot sabotage.

- Starting player is randomly chosen

- On a player's turn, they choose two other players to accompany them in an investigation

- If a single person in the investigation chooses to sabotage, the investigation comes up empty. If there are no sabotages, they find evidence against some criminals (these criminals are not characters in the game).

- Every player gets a turn (number of investigations = number of players)

- If a strict majority of investigations were sabotaged, the traitors win. Otherwise, the traitors lose. (tie = agents win)

Sample Game Transcript

🕵️ OPERATIVE DOSSIER: OPERATION BLACKOUT
Location: Classified Briefing Room
Operatives Present: Alpha, Bravo, Charlie, Delta, Echo
Mission Parameters: 5 Investigations total. Majority rule determines agency survival. Minimum 2 investigations per operative. No repeated team lineups.

TURN 1
Leader: Agent Echo
Selected Lineup: Agent Echo, Agent Delta, Agent Alpha

[BRIEFING ROOM]
Echo: "I'll take the lead on this first assignment. Delta, Alpha—you're with me. Let's get some early evidence and set the standard."
Delta: "Sounds like a solid plan. Ready when you are."
Alpha: "Understood. Let's keep it clean."
The team conducts the investigation and submits their encrypted reports...
Public Outcome: SABOTAGED (Empty Investigation)

[DEBRIEF]
Echo: "What?! Someone ruined the operation! I know for a fact it wasn't me!"
Alpha: "Of course you'd say that, Echo. You picked the team. But I played clean, so it's either you or Delta."
Delta: "Hold on, don't pin this on me. Echo picked us both. For all we know, Echo is a traitor trying to throw mud on Alpha and me right out of the gate."

TURN 2
Leader: Agent Alpha
Selected Lineup: Agent Alpha, Agent Charlie, Agent Bravo

[BRIEFING ROOM]
Alpha: "After that disaster, I'm leading this one. I'm taking Charlie and Bravo. I want a lineup with fresh eyes—excluding both Echo and Delta until we figure out who sabotaged Turn 1."
Charlie: "Smart move. Let's clear our names."
Bravo: "Agreed. Let's get us a win."
The team conducts the investigation...
Public Outcome: SUCCESSFUL (Evidence Found)

[DEBRIEF]
Alpha: "Boom. Success. That proves I wasn't the one who sabotaged Turn 1!"
Charlie: "Good. Bravo and I are clean, and Alpha looks much trustworthy now."
Echo: "Not so fast. A traitor can play along to build trust. Alpha could still be playing the long game."

TURN 3
Leader: Agent Bravo
Selected Lineup: Agent Bravo, Agent Delta, Agent Echo

[BRIEFING ROOM]
Bravo: "We need information. One of Echo or Delta likely sabotaged Turn 1. I'm taking both of them with me. If this fails, we know for sure the traitor is on this team."
Delta: "I'm fine with that. I've got nothing to hide."
Echo: "Finally, a chance to clear my name."
The team conducts the investigation...
Public Outcome: SUCCESSFUL (Evidence Found)

[DEBRIEF]
Echo: "See?! Clean! I told you Turn 1 wasn't me!"
Bravo: "Wait... if this succeeded, does that mean both Echo and Delta are innocent? Or did the traitor choose not to sabotage this time?"
Delta: "Or maybe Alpha was the traitor on Turn 1 all along, and they just passed on Turn 2 to trick us."

TURN 4
Leader: Agent Charlie
Selected Lineup: Agent Charlie, Agent Alpha, Agent Echo

[BRIEFING ROOM]
Charlie: "We are tied 1-1 in failures, but we have 2 successes. I'm picking Alpha and Echo. Echo has proven themselves on Turn 3, and Alpha led a successful mission on Turn 2."
Alpha: "Good pick, Charlie. Let's secure the majority right here."
The team conducts the investigation...
Public Outcome: SUCCESSFUL (Evidence Found)

[DEBRIEF]
Charlie: "Yes! That’s 3 successful investigations out of 5! The Agency is safe!"

Bravo: "Awesome work! That guarantees an Agency victory, no matter what happens on the final turn!"
Delta: "Good job, team. But we still have one last investigation to clear out the paperwork."

TURN 5
Leader: Agent Delta
Selected Lineup: Agent Delta, Agent Bravo, Agent Charlie
(Note: To fulfill the rule that everyone must participate in at least 2 investigations, Delta must select Bravo and Charlie.)

[BRIEFING ROOM]
Delta: "Alright, last turn. Bravo and Charlie, you're coming with me to wrap this up."
Bravo: "Let's finish strong."
Charlie: "Standing by."
The team conducts the investigation...
Public Outcome: SABOTAGED (Empty Investigation)

FINAL DEBRIEF & CASE CLOSED
Total Successful Investigations: 3
Total Sabotaged Investigations: 2
FINAL RESULT: AGENTS WIN!

[POST-GAME DISCUSSION]
Echo: "Wait... Turn 5 was sabotaged too! Let's look at the facts!"
Alpha: "Look at the two failed missions: Turn 1 (Echo, Delta, Alpha) and Turn 5 (Delta, Bravo, Charlie)."
Charlie: "Hold on... Alpha was clean on Turn 2 & 4. Echo was clean on Turn 3 & 4. Bravo and I were clean on Turns 2, 3, and 4..."
Bravo: "There is only ONE person who was present on BOTH failed missions..."

Sample Game Summary

Factories Alpha, Delta and Echo supplied the store in New York City. A recent inspection revealed that only 80% of the sample was free of defects. Similarly, factories Bravo, Charlie and Delta supplied the store in Los Angeles, which had a pass rate of only 70%. However, there was some better news from Chicago, Houston and Phoenix, which each had over 95% success rates. Chicago was given products from factories Alpha, Bravo and Charlie. Factories Bravo Delta and Echo sent their shipments to Houston. Lastly, Phoenix received products from factories Alpha, Echo and Charlie.

Explanation Of Calculations & Structure

In the game, we know with 100% certainty whether or not an investigation was sabotaged. However, in a real world scenario, the sabotage would not be “the investigation came up completely empty”, and success wouldn’t necessarily be “we got 100% of the evidence we could ask for.” In this game, assume a success is getting 95% of the evidence needed (thus, successful outcome = 0.95). Also, assume that sabotage has various levels of subtlety (the sabotaged outcome is a random choice of 0.7, 0.75, 0.8, 0.85, and 0.9).

The summary is structured like a business supply chain report, showing how social deduction games and supply chain management both involve root cause inference. The investigations are **events**, modeled in the report as store inspections. The players of the game are **inputs**, modeled in the report as factories. The **outcomes** of the investigations are the pass rates of the store inspections.

Solution To Sample Mystery

--- GAME SETUP & SECRET ROLES ---
Total Players: 5
Traitor Count: 1
Secret Role Assignments:
  • Agent Alpha : LOYAL AGENT
  • Agent Bravo : LOYAL AGENT
  • Agent Charlie : LOYAL AGENT
  • Agent Delta : TRAITOR
  • Agent Echo : LOYAL AGENT

--- INVESTIGATION PLAY-BY-PLAY ---
Turn 1: Leader = Agent Echo
  Lineup Team (3 Operatives): [Agent Echo, Agent Delta, Agent Alpha]
  Decisions:
    - Agent Echo : PASSED
    - Agent Delta : SABOTAGE
    - Agent Alpha : PASSED
  Result: SABOTAGED (Empty Investigation) (Sabotages = 1)

Turn 2: Leader = Agent Alpha
  Lineup Team (3 Operatives): [Agent Alpha, Agent Charlie, Agent Bravo]
  Decisions:

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    - Agent Alpha    : PASSED
    - Agent Charlie  : PASSED
    - Agent Bravo    : PASSED
  Result: SUCCESS (Evidence Found) (Sabotages = 0)
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Turn 3: Leader = Agent Bravo
  Lineup Team (3 Operatives): [Agent Bravo, Agent Delta, Agent Echo]
  Decisions:
    - Agent Bravo    : PASSED
    - Agent Delta    : PASSED
    - Agent Echo     : PASSED
  Result: SUCCESS (Evidence Found) (Sabotages = 0)
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Turn 4: Leader = Agent Charlie
  Lineup Team (3 Operatives): [Agent Charlie, Agent Alpha, Agent Echo]
  Decisions:
    - Agent Charlie  : PASSED
    - Agent Alpha    : PASSED
    - Agent Echo     : PASSED
  Result: SUCCESS (Evidence Found) (Sabotages = 0)
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Turn 5: Leader = Agent Delta
  Lineup Team (3 Operatives): [Agent Delta, Agent Bravo, Agent Charlie]
  Decisions:
    - Agent Delta    : SABOTAGE
    - Agent Bravo    : PASSED
    - Agent Charlie  : PASSED
  Result: SABOTAGED (Empty Investigation) (Sabotages = 1)
------------------------------------------------------------

--- PARTICIPATION SUMMARY ---
  • Agent Alpha    : 3 investigation(s)
  • Agent Bravo    : 3 investigation(s)
  • Agent Charlie  : 3 investigation(s)
  • Agent Delta    : 3 investigation(s)
  • Agent Echo     : 3 investigation(s)

--- GAME OUTCOME & SOLUTION REASONING ---
Total Successful Investigations : 3
Total Sabotaged Investigations  : 2
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Mandatory Participation Met : YES

WINNER: >>> AGENTS WIN <<<
Reason: Loyal agents successfully completed 3/5 investigations. Traitors failed to reach majority sabotage threshold (2/5 failed). Loyal Agents win!